James Scholar Project Update

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1. K-means on my descriptions

I ran k-means ($k = 2$) on the 256 training descriptions without using the labels, then compared the clusters with L1–L5.

	L1	L2	L3	L4	L5
Cluster A (autonomous, robotic, lab, materials)	7	7	23	41	40
Cluster B (learning, deep, neural, data)	90	30	11	6	1

The text mostly splits into ML methods vs. autonomous labs, so low and high levels largely separate on their own. L3 is split across both clusters, which matches L3 being the hardest class for the classifiers. With $k = 5$, the clusters follow topic rather than level (adjusted Rand index 0.19).

2. Data for the independent test

Sheet in the newest inventory	New projects	Levels
AI Systems & Labs (same type as my data)	14	L1–L5
Models, Tools & Software	260	almost all L1–L2

Only 14 new projects are the same type as my training data, well short of 60.

Questions

1. Should the clustering go into the final report? Is the adjusted Rand index a good way to compare clusters with labels?
2. If I still have fewer than 60 new projects at the end of the semester, should I report the small test with a confidence interval, add the Models/Tools sheet as a separate test, or both?